

Corrected shared-contact shaking: accuracy and cost

The prospectively declared source is `38f407d12208654075c07315e96b9fc213612b91`. All nine attempts completed; both fixed-step refinement edges qualify, every timed repetition passes physical/residual gates and full trajectory budgets, and repeated histories are bitwise identical.

The scene contains 27 one-kilogram spheres in a container prescribed to shake at plus/minus 20 m/s for 0.12 s, with gravity 9.81 m/s squared, pair friction 0.4 and zero restitution. The common-point convention is applied to complete contact rows, warm angular impulses and boundary work. Parameters are synthetic.

Setting	Native seconds, all repetitions	Median seconds
Fixed 1.25 microsecond reference	22.768746, 23.051115, 19.541006	22.768746
Motion-based travel guard 0.015	3.261182, 3.173906, 3.051779	3.173906

The native median ratio is **7.17 times**. Native timing includes stepping/state recording; whole-process timings are retained separately. Sequential interleaved launches reduce ordering bias but external machine load is uncontrolled.

Fixed-step refinement	Position RMS, m	Velocity RMS, m/s	Spin RMS, rad/s	Orientation RMS, rad
fixed_5us- >fixed_2_5us	8.3173862e-06	0.00056231781	0.015178209	0.00034093656
fixed_2_5us- >fixed_1_25us	6.1742988e-06	0.00052242843	0.010987465	0.00023586459

Candidate versus finest reference: position 2.2303661e-05 m, velocity 0.0015628478 m/s, spin 0.052144196 rad/s, orientation 0.0011607172 rad.

Full budgets: 5 mm, 0.05 m/s, 0.1 rad/s and 0.01 rad; reference edges use one quarter. These are declared error tolerances, not a demonstrated asymptotic convergence order. This result concerns one synthetic scene and horizon; it does not qualify random hulls, authentic rubber, general CCD, online error control or Vektor-native execution. The earlier 6.11-times separate-endpoint result is preserved as historical evidence.

Audit: `python -m research.audit_shared_shake`. Exact source, plan, binary hash, all raw histories and individual times are retained in `results/`.